Abdullah Ejaz

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SUMMARY

Computer Science undergraduate with hands-on experience in full-stack web development and applied machine learning. Skilled in building and deploying end-to-end ML pipelines, distributed systems, and production web applications targeting ML/AI and full-stack engineering roles.

EDUCATION

National University of Sciences and Technology (NUST) — Islamabad, Pakistan *2023 – Present*
BS Computer Science (6th Semester) | CGPA: **3.66 / 4.0**

EXPERIENCE

DevDen — Programming Intern *June 2022 – August 2022*
Faisalabad, Pakistan

- Built and deployed **React.js** front-end modules for a production website, translating Figma wireframes into responsive, pixel-perfect UI components.

PROJECTS

NastaRead — Distributed Urdu OCR @ Scale — *Python, PyTorch, Keras, Hadoop MapReduce, Docker, FastAPI* *2026*
End-to-end pipeline to restore and digitize degraded Nastaleeq-script Urdu documents at scale.

- Trained an **SMP U-Net** image restoration model and a **7-block Conv-Transformer** OCR model achieving minimal character error rate (~8% CER) on the MMU-OCR-21 and NUST-UHWR benchmarks.
- Built a degradation pipeline (Gaussian blur, salt-and-pepper noise, skew, contrast reduction) to artificially augment training data for robustness.
- Parallelized inference using **Hadoop MapReduce** in a fully containerized Docker environment, enabling horizontal scaling for large document batches.
- Deployed as a live web demo via **FastAPI**, processing full document restoration and OCR end-to-end through a REST API.

SmogPenalty — Solar Power Loss Predictor — *TypeScript, Next.js, FastAPI, Python, Scikit-Learn, LGBM, Plotly* *2025*
Full-stack ML web app predicting solar power output and weather-driven losses for any GPS location.

- Designed a **hybrid ML model** combining LightGBM and log-transformed Linear Regression, trained on 3 years of multi-grid solar and climate data from the US and Australia.
- Integrated **NASA POWER API** and **OpenAQ API** to supply real-time irradiance, temperature, and air quality data as model features at inference time.
- Built an interactive **Next.js** frontend with Plotly visualizations for per-location power generation and loss breakdowns; backend served via **FastAPI**.

CrimeVision — Islamabad Crime Analysis Platform — *Node.js, Express.js, PostgreSQL, MongoDB, Redis, Next.js* *2025*
City-wide full-stack crime analysis and visualization platform. Live at crimevision.live.

- Architected and implemented a **polyglot backend** using Node.js/Express with **PostgreSQL (Supabase)** for relational data, **MongoDB Atlas** for unstructured records, and **Redis (Upstash)** for caching.
- Designed a **normalized relational schema** in Supabase and integrated **Supabase Auth** for user authentication and row-level security.
- Deployed frontend on **Vercel** and backend on **Render**; configured a custom domain via name.com for a production-ready public release.

SKILLS & TECHNOLOGIES

Languages & Frameworks: Python, C/C++, Java, TypeScript, JavaScript, Next.js, React.js, Node.js, Express.js, FastAPI
ML / AI: Scikit-Learn, PyTorch, TensorFlow/Keras, model training & evaluation, data preprocessing & visualization (Pandas, Plotly, Matplotlib & Relevant Libraries), n8n, make.com/zapier
Databases: PostgreSQL, MongoDB, MySQL, Redis, Supabase
Infrastructure & Tools: Docker, Hadoop MapReduce, Git/GitHub, Vercel, Render, Postman, Figma, Photoshop, Copilot, Codex
Design: UI/UX Design, Wireframing & Prototyping (Figma), User Testing (Maze)